

# Improved Rain Mapping by Graph Signal Processing Tools

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**Abstract**—In this paper, we apply, for the first time, the use of graph signal processing (GSP) to the emerging application of near-ground two-dimensional rainfall mapping based on signal level measurements from commercial microwave link (CML). Most existing rainfall reconstruction algorithms are based on spatial interpolation of assumed point rain measurements, arbitrarily representing each CML by a virtual rain gauge (VRG), located at the center of the CML. To demonstrate the usability of GSP tools in this setting, we challenge the arbitrary location of the VRG, looking for the best location of the VRG along the CML in some sense. Here, we show that using GSP tools enables the identification of alternative representative locations along the links that yield improved rainfall maps. The proposed approach is validated using real CML data, demonstrating its effectiveness in enhancing map accuracy.

**Index Terms**—Graph Signal Processing (GSP), Commercial Microwave Links (CMLs), rainfall mapping, inverse problem, environmental monitoring, inverse distance weighting (IDW)

## I. INTRODUCTION

Opportunistic sensing is especially valuable in rainfall monitoring, where commercial microwave link (CML) complements gauges and radar by helping close coverage gaps and providing higher resolution insights [1], [2]. By capturing local variations that other instruments may overlook, CMLs can enhance flood warning systems, support agricultural decision making, and improve the overall reliability of weather monitoring [3].

Many complex systems in engineering, physics, and biology consist of networks of interacting units that give rise to graph signals [4]. The emerging field of graph signal processing (GSP) extends concepts and techniques from traditional digital signal processing (DSP) to data indexed by generic graphs [4]–[6]. However, most existing tools assume stationary topologies of the underlying graph and track signals rather than the graphs themselves.

Various studies have shown over the last years the potential of CMLs information for rainfall monitoring and 2-D reconstruction. One common approach is to treat each link as a virtual rain gauge (VRG) sensor located at its midpoint [7]. Although this approximation yields reasonable performance, it neglects the fact that rainfall is inherently heterogeneous along the link path. Since CMLs provide path-integrated measurements rather than point observations, assuming a fixed midpoint representation may lead to suboptimal performance, limiting the accuracy of high-resolution mapping.

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We claim that there exists a virtual point  $x^* \in [0, L]$  along each link that better represents the rainfall along the path compared to the center of the CML. Once this point is identified, existing interpolation methods, such as inverse-distance weighting (IDW) and kriging, can be used to improve rain mapping accuracy. Although we are limited to only the CML geometry and aggregated attenuation measurements, while the pointwise rainfall profile  $R(\mathbf{x})$  where  $\mathbf{x} = (x, y)$  remains unknown [8], we derive a new method that identifies a representative point along each link, combining CML data with GSP techniques [9]. This approach addresses the challenge by exploiting the spatial smoothness of rainfall fields and framing the task as an inverse problem: given the link geometries and path-average rainfall estimates  $\hat{r}$ , it searches for representative points, each with its rainfall value, such that the reconstructed graph signal is smoother in the GSP sense [10], meaning consistent with the CML measurements [11].

Thus, our main contribution is a new method for improving rainfall map reconstruction. The proposed method outputs representative points along CMLs that minimize the graph smoothness of the path-averaged rainfall estimates under a coverage constraint. To demonstrate that these representative points outperform the conventional midpoint assumption, we employ a standard spatial interpolation method to generate rainfall maps and compare the resulting reconstruction accuracy.

The remainder of this paper is organized as follows. Sec. II provides the necessary background on GSP and rainfall field reconstruction. Sec. III describes the proposed model formulation, including the graph construction and the Improved Rainfall field Reconstruction Algorithm. Sec. IV presents the experiments and results on real data to demonstrate the effectiveness of the proposed approach. Finally, Sec. V concludes the paper and discusses potential directions for future work.

## II. GSP AND RAINFALL MODELING BACKGROUND

### A. GSP Background

GSP is a framework for analyzing data by extending classical DSP concepts to signals defined over irregular domains, represented by graphs [4], [5]. A weighted undirected graph  $\mathcal{G}(\mathcal{V}, \xi, \mathbf{W})$  consists of a set  $\mathcal{V}$  of  $N = |\mathcal{V}|$  nodes (or vertices), set of edges  $\xi$ , and  $\mathbf{W} \in \mathbb{R}^{N \times N}$  is the weighted adjacency matrix with nonnegative entries  $\mathbf{W}_{n,m}$  representing the similarity between nodes  $n$  and  $m$ . The degree matrix  $\mathbf{D}$  is defined as the diagonal matrix with the entries

$$\mathbf{D}_{n,n} = \sum_{m=1}^N \mathbf{W}_{n,m}, \quad n = 1, \dots, N, \quad (1)$$

and the combinatorial graph Laplacian matrix is given by  $\mathbf{L} = \mathbf{D} - \mathbf{W}$ . The Laplacian plays a role analogous to the frequency operator in classical signal processing. It enables a notion of the graph Fourier transform, which is the basis for various GSP tools [12].

A graph signal is a vector  $\mathbf{x} \in \mathbb{R}^N$  whose entry  $x_n$  represents the signal value at node  $n$ . The *smoothness* of graph signals quantifies how similar signal values are across connected nodes [6]. A common smoothness measure is the Dirichlet energy

$$\text{SMO}(\mathbf{x}) = \mathbf{x}^T \mathbf{L} \mathbf{x} = \frac{1}{2} \sum_{n=1}^N \sum_{m=1}^N \mathbf{W}_{n,m} (x_n - x_m)^2. \quad (2)$$

Here, each squared difference  $(x_n - x_m)^2$  is weighted by the edge weight  $\mathbf{W}_{n,m}$ , so discrepancies between strongly connected nodes are penalized more than those between weakly connected ones. Thus, small values of  $\text{SMO}(\mathbf{x})$  indicate smooth variation of the signal over the graph, consistent with the spatial coherence of natural fields such as rainfall.

### B. Rainfall Field Reconstruction

CMLs, widely deployed in cellular backhaul networks, provide a cost-effective alternative for rainfall monitoring by exploiting the attenuation of radio signals caused by rain. Unlike traditional instruments, CMLs measure path-integrated attenuation at high spatial density, which can later be translated into rain estimates using the power-law equation [2].

For a link of length  $L$ , the measured rain-induced path-integrated attenuation  $A$  (dB) can be approximated by:

$$A = \int_0^L a [R(s)]^b ds. \quad (3)$$

The parameters  $a$  and  $b$  depend mainly on the frequency, polarization, and raindrop size distribution. By inverting this relation, the measured attenuation from each CML can be converted into a path-averaged rainfall estimate  $\hat{r}$ , which serves as a virtual gauge value associated with the link:

$$\hat{r} = \left( \frac{1}{L} \int_0^L R(x)^b dx \right)^{1/b}. \quad (4)$$

The rainfall field at any point  $\mathbf{x} = (x, y)$ , denoted by  $\hat{R}(\mathbf{x})$ , is reconstructed using IDW from the set of  $N$  VRGs obtained from  $N$  CMLs [13].

## III. MODEL FORMULATION

In this section, we focus on representing each CML by a single point by exploiting the empirical spatial coherence of rainfall. Accordingly, we first introduce how to connect CMLs to the baseline graph (see Subsection III-A). Then, we show how to determine the maximum coverage area for a specific topology (see Subsection III-B). Using this information, we present the smoothness model in Subsection III-C, followed by the full algorithm that minimizes graph smoothness subject to a coverage constraint in Subsection III-D.

### A. Baseline Graph

In this section, we present the baseline graph construction, which will be updated later in Subsections III-C and III-D.

We represent the rainfall field recovery problem as a graph [14], where each CML is mapped into the graph domain to capture the spatial connections of rainfall.

**Nodes:** Each CML is associated with a VRG located at its midpoint, which provides a rainfall measurement derived from the link attenuation and length. Thus, for  $N$  links, we obtain  $N$  nodes representing rainfall measurements.

**Edges and Weights:** The nodes are connected based on geographic proximity, reflecting spatial relationships between rainfall estimates. We use a distance-decay weight with cutoff  $D_{\max}$  such that

$$W_{m,n} = \begin{cases} \frac{\alpha}{d_{mn}^2}, & \text{if } d_{mn} \leq D_{\max}, \\ 0, & \text{if } d_{mn} > D_{\max}, \end{cases} \quad (5)$$

Where  $W_{m,n}$  is the weight between nodes  $m$  and  $n$ ,  $d_{mn}$  is the geographic distance between their midpoints,  $\alpha$  is a scaling constant, and  $D_{\max}$  is a parameter that defines the maximum range of connectivity. The graph Laplacian is then obtained as described in Subsection II-A. This construction enforces local spatial coherence while avoiding spurious long-range connections.

Fig. 1 illustrates the process of converting link midpoints into nodes and forming a graph representation.

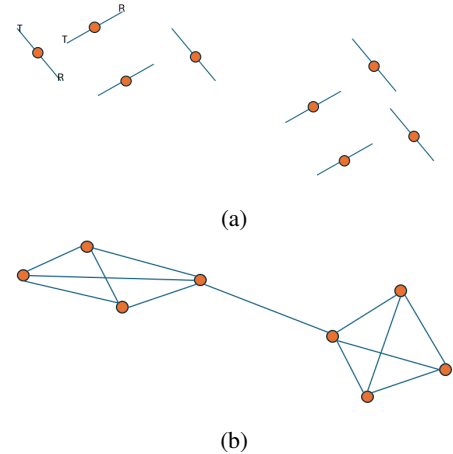


Fig. 1: Example of CMLs in geographic space with nodes at link midpoints (a) and the corresponding graph representation (b).

### B. Coverage-Based Optimization of Node Locations

To ensure spatial representativeness, we evaluate the coverage provided by the selected representative points. The idea is to maximize the total area covered by a given map using a specified topology. By adjusting the position of a node along a link, we select the configuration that achieves the highest coverage score.

Let  $C$  denote the radius of influence associated with each point, and let  $N$  be the number of links. The overall coverage is defined as

$$\text{COV} = \max_{(x_1, y_1) \in \text{link}_1, \dots, (x_N, y_N) \in \text{link}_N} \left| \bigcup_{k=1}^N B_C(x_k, y_k) \right|, \quad (6)$$

Here, COV is the total area of the union of coverage circles centered at the selected points. Each  $B_C(x_k, y_k)$  is a circle of radius  $C$  centered at the selected representative point of link  $k$ . This formulation favors configurations where the influence regions overlap minimally, thereby maximizing the effective coverage area of the map. Fig. 2 shows optimal node positions (in red) that maximize the covered area for a given  $C$  and network topology.

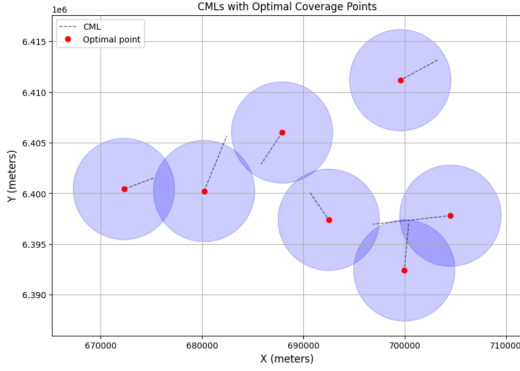


Fig. 2: Coverage-Based Optimization of Node Locations.

### C. Smoothness-Based Optimization of Node Locations

To refine the representative point selection, we exploit the structural property of smoothness of the rainfall field over the constructed graph. Let  $\hat{\mathbf{r}}$  denote the vector of all VRG measurements, and let the vector  $\mathbf{x} = \{(x_n, y_n)\}_{n=1}^N$  represent the spatial coordinates of the  $N$  graph nodes. Then, the smoothness of this graph  $G$ , with  $N$  nodes, can be quantified by the normalized Dirichlet energy [6]:

$$\text{SMO}(G) = \frac{1}{\lambda_{\text{avg}}} \frac{\hat{\mathbf{r}}^T \mathbf{L} \hat{\mathbf{r}}}{\|\hat{\mathbf{r}}\|^2}, \quad (7)$$

where  $\mathbf{L}$  is the graph Laplacian obtained in Subsection III-A and  $\lambda_{\text{avg}}$  is the average eigenvalue of  $\mathbf{L}$ . This measure penalizes

discrepancies between strongly connected nodes, ensuring that rainfall estimates at nearby locations remain consistent.

Therefore, we aim to find the smoothest graph. This ensures that the chosen positions best reflect the spatial relationships implied by the graph and are the most likely locations where the actual rainfall occurred. Formally, the optimization problem is  $\mathbf{x}^* = \arg \min_{\mathbf{x}} \text{SMO}(G)$ , where  $\mathbf{x}^*$  are the optimal node locations that minimize the smoothness measure.

### D. Improved Rainfall Field Reconstruction Algorithm

We want to construct the most accurate rainfall field map by combining the **Coverage-Based** and **Smoothness-Based** Optimizations of Node Locations. In this way, we minimize the smoothness term to ensure that the resulting graph is as consistent with the original rain as possible, while preserving coverage across the map. We then solve the following optimization problem:

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} \text{SMO}(G) \quad \text{s.t.} \quad \frac{\text{COV}}{\bigcup_{k=1}^N B_C(\mathbf{x}_k^*)} \leq \tau. \quad (8)$$

Here,  $\text{SMO}(G)$  quantifies the correlation between rainfall values at the nodes and the edge weights, while  $\tau$  is a threshold target parameter that ensures coverage remains above a fraction of the maximum attainable coverage. The optimization problem is non-convex and is solved using a discrete sequential (greedy) search. Each link is uniformly sampled, and candidates are evaluated iteratively by selecting the point that minimizes the smoothness objective while satisfying the coverage constraint. The full algorithm flow is shown in Fig. 3.

### E. Discussion

We have a few parameters that can be tuned to influence the performance of the system or serve as targets:

- $\tau = \frac{\text{COV}}{\bigcup_{k=1}^N B_C(\mathbf{x}_k^*)}$ : This parameter can be used as a target criterion, for example by aiming to achieve 90% of the maximum attainable coverage.
- $C$  — radius of influence: This parameter can be controlled and is typically set based on the approximate size of a rain cell, which is about 5 km.
- $D_{\text{max}}$  — maximum connection distance in the graph: This can be set either as a fixed constant that represents twice the rain cell size (e.g., 10 km). This parameter can also be tuned for each topology in future research.

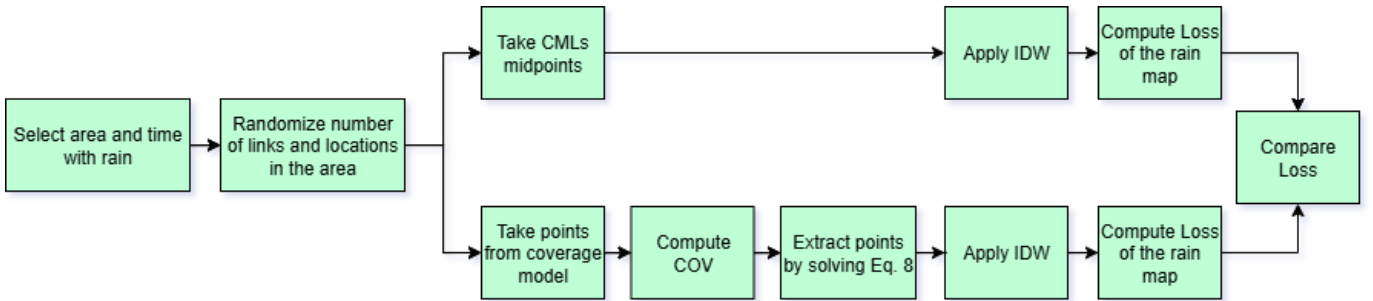


Fig. 3: Algorithm flow diagram. The upper part shows the baseline, while the lower part shows the smoothness–coverage algorithm.

## IV. EXPERIMENTS

### A. Setup

We demonstrate our results using real data from the OpenMRG dataset [15], which includes 364 bidirectional links, along with radar observations provided on a  $1 \times 1$  km spatial grid. CML data were sampled every minute, while radar data were available every five minutes and aggregated into hourly sums. We restrict the network to links longer than 3 km located within the radar and gauge validation areas. The CML frequency range spans 7.5–38.5 GHz, with a median of 28.2 GHz. The full setup is shown in Fig. 4.

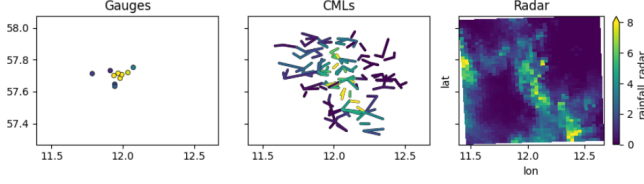


Fig. 4: OpenMRG dataset, including gauges, radar, and CMLs.

### B. Methods of comparison

We produce two gridded rainfall maps using the same IDW interpolation and compare both to the radar reference. First, the points along each link where the VRG is located are derived using our proposed algorithm and then interpolated to the radar grid with IDW. Second, each link is represented by a VRG at its geometric midpoint and interpolated with the same IDW settings to produce the naive midpoint map. Radar and CML data are aggregated to the same hourly totals and aligned on the same grid; actual gauges are used only as a sanity check and are not used for tuning the interpolation. Comparisons are computed between each of the two maps and the radar map and evaluated using the structural similarity index (SSIM); the method with the higher SSIM performs better [16].

### C. Results

In the first scenario, this experiment was applied to all 135 links in the study area after filtering for this study, using the following parameters:  $\tau = 1.1$ ,  $C = 5$  km, and  $D_{\max} = 10$  km. This comparison shows a 5% improvement in the SSIM metric when using the algorithm's points.

In the second scenario, we show one permutation of the links, comprising 85 out of the 135 links, with both the selection and combination randomized. We then apply the algorithm with the same parameters as in scenario 1 and present the algorithm-selected points in Fig. 5, along with the corresponding IDW maps for the midpoints and for the algorithm's points in Fig. 6. The algorithm's points yield a 13.5% improvement in SSIM compared to the midpoint method.

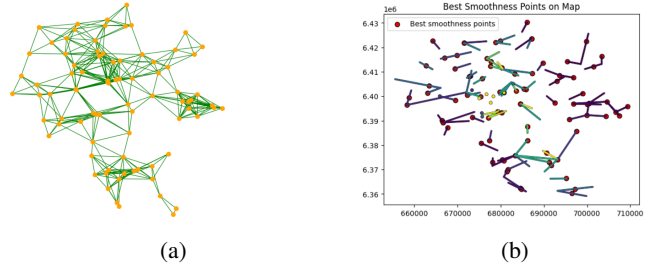


Fig. 5: Algorithm's final graph (a), selected points along each link (b).

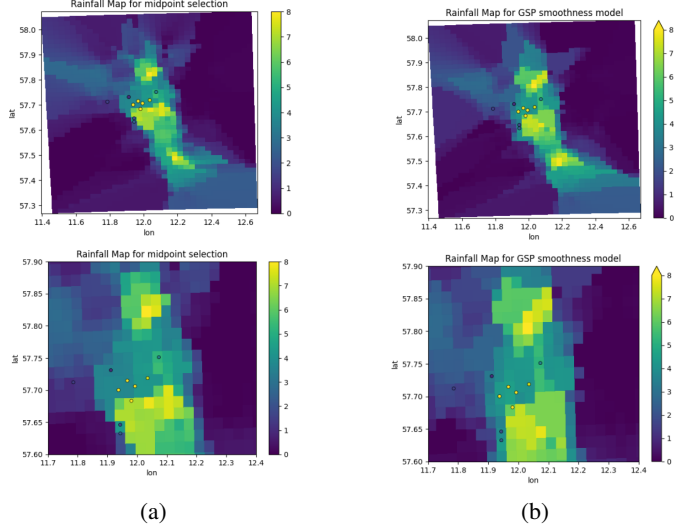


Fig. 6: IDW interpolation using midpoint (a) vs. algorithm-selected points (b). Upper images show full view, bottom images are zoomed-in.

## V. CONCLUSION

In this work, we introduced a GSP-based framework to improve rainfall mapping from CML measurements by selecting optimal representative points along each link. By leveraging the spatial smoothness of rainfall fields, the proposed method enhances reconstruction accuracy, particularly in scenarios with sparse link availability. The results show that graph-based smoothing improves performance across different rainfall intensities, with the largest gains observed in regions with limited measurement coverage. This demonstrates the ability of the proposed approach to exploit spatial correlations and mitigate information gaps in the network. Rain map reconstruction can be further improved through adaptive parameter tuning. In particular, overly large values of  $D_{\max}$  may degrade performance, while very small values reduce the influence of graph connectivity. Optimizing this parameter for a given topology represents an important direction for future work.

Overall, the proposed method provides representative points along each link that act as VRGs, enabling the construction of rainfall maps using spatial interpolation techniques like IDW that more closely align with radar observations. Beyond rainfall estimation, this framework offers potential for applying additional GSP tools to CML data or other environmental field reconstruction involving irregularly deployed measurements.

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